# Experiment Targets

Planning guide for *After the Decision*

These are **illustrative research goals, not forecasts or observations**. Recorded comparators remain unchanged. Blue T rows show an arithmetically consistent target scenario, normalized to existing denominators. Statistical success and generalization still require real evaluation.

| Group | Body output | Research question | Goal checkpoint (T) |
|---|---|---|---|
| E1 | Table 3 / Fig. 3 | Natural gain and delayed recovery | $+1.02$ / $+1.23$ pp* |
| E2 | Table 4 | Correct source attribution | Local beats permutation |
| E3 | Table 5 | Learnable cold-start headroom | 81.86 / 81.72% |
| E4 | Table 6 | Value of the update alone | $+1.39$ / $+1.20$ pp* |
| E5 | Table 7 | Cost at one call per input | Selection $\leq 30/35\,\mu$s |
| E6 | Table 8 | Refusal at matched coverage | 95.78%; $U = -0.0845$ |

* Natural / controlled gain versus current Relay in E1, current corrected Query in E4. These E-labels match the presentation draft; the active standard's E6 means untouched confirmation.

## 1. Primary selection and delay

Four providers; 157 molecules / 124 scaffold clusters; $D \in \{0, 1, 4\}$. Each method has 4,710 calls across 30 schedules, *not* 4,710 independent molecules. Controlled drift is an evaluator-side label flip after tick 78.

**Table 1:** E1 / body Table 3. Recorded anchors and planning target (T).

| | Natural menu | | Controlled drift | |
|---|---|---|---|---|
| Method / status | Correct | Success (%) | Correct | Success (%) |
| Random (M) | 3,945 | 83.76 | — | — |
| Best Fixed (M) | 4,152 | 88.15 | — | — |
| Frozen (M) | 4,230 | 89.81 | 3,456 | 73.38 |
| MIRA-Relative (M) | 4,230 | 89.81 | — | — |
| Online SGD (M) | 4,202 | 89.21 | 3,958 | 84.03 |
| Greedy LinUCB (M) | 4,226 | 89.72 | 3,822 | 81.15 |
| BayesianRouter (M) | 4,217 | 89.53 | 3,781 | 80.28 |
| Current MIRA-Relay (M) | 4,230 | 89.81 | 4,124 | 87.56 |
| Retrospective Best Fixed† (M) | 4,260 | 90.45 | — | — |
| Improved MIRA-Relay (T) | 4,278 | 90.83 | 4,182 | 88.79 |

† Retrospective Best Fixed chooses one fixed provider with evaluator truth; it is not an upper bound on contextual selection. The natural per-molecule oracle is 150/157 = 95.54%, or 4,500 recorded decisions. T adds 48 correct natural decisions: 17.8% of Frozen's 270 recoverable decisions.

**Table 2:** E1 / body Figure 3 data. Gains over Frozen, in pp; T is illustrative.

| | Natural menu | | Controlled drift | |
|---|---|---|---|---|
| Delay | Current (M) | Target (T) | Current (M) | Target (T) |
| 0 | $-0.13$ | $+1.15$ | $+13.63$ | $+15.67$ |
| 1 | $-0.13$ | $+1.02$ | $+14.52$ | $+15.41$ |
| 4 | $+0.25$ | $+0.89$ | $+14.39$ | $+15.16$ |

Each delay has 1,570 calls. T correct counts: natural 1,428/1,426/1,424; controlled 1,398/1,394/1,390. Their sums reproduce Table 1. A compatible aggregate rescue/harm scenario is 110/62 (natural) and 818/92 (controlled), giving net 48 and 726 versus Frozen. This is not a predicted event trace.

**Decision test.** Seek a practically meaningful natural gain and retain controlled recovery; require prospectively specified paired scaffold uncertainty against the strongest legal matched control. Target totals alone cannot demonstrate significance. Delay points are goals, not a fitted law.

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## 2. Mechanism and learnable headroom

E2 preserves initial scores, source residual computation, budget and release clock; only the credit destination changes. Its scaffold-equal estimand differs from E1's pooled point estimates.

**Table 3:** E2 / body Table 4. Source-credit contrasts (pp); goals are dependent.

| | Natural | | Controlled | |
|---|---|---|---|---|
| Contrast | Current (M) | Target (T) | Current (M) | Target (T) |
| Local − Frozen | +0.10 | +1.10 | +14.40 | +15.50 |
| Permutation − Frozen | +0.41 | +0.10 | −1.27 | −1.00 |
| Local − permutation | −0.30 | +1.00 | +15.67 | +16.50 |

For T: $1.10 - 0.10 = 1.00$ and $15.50 - (-1.00) = 16.50$. The three contrasts are not independent successes. Measured contrasts use unrounded values before subtraction. Actual paired intervals must support the attribution claim; do not derive intervals from these target means.

**Table 4:** E3 / body Table 5. Static cold-start targets on 722 development molecules.

| Selector / status | Correct / 722 | Accuracy (%) |
|---|---|---|
| Best Fixed (M) | 572 | 79.22 |
| Raw Frozen (M) | 573 | 79.36 |
| Corrected Frozen (M) | 579 | 80.19 |
| BCE + CLS (M) | 567 / 570 | 78.53 / 78.95 |
| BT + CLS (M) | 569 / 565 | 78.81 / 78.25 |
| Improved prior (T) | 591 / 590 | 81.86 / 81.72 |
| Per-molecule oracle† (M) | 618 | 85.60 |

Paired seeds 0/1 use the same molecules; '/' separates seed results. The oracle is evaluator-only. Of 722 molecules, 471 have all providers correct, 147 are mixed and 104 are all wrong. T captures 12/11 of the 39 remaining selection opportunities (30.8/28.2%). It needs a better ranking representation or training rule; current learned priors have not demonstrated this effect.

**Decision test.** Both seeds should exceed the corrected Frozen by at least 1 pp, without negative leave-one-fold-out contrasts. Static ranking success alone does not prove online adaptation.

## 3. Isolate the update

**Table 5:** E4 / body Table 6. Score × update diagnostic; 2,166 calls per cell.

| | Natural | | Controlled | |
|---|---|---|---|---|
| Score / update / status | Correct | Accuracy (%) | Correct | Accuracy (%) |
| Raw / Frozen (M) | 1,719 | 79.36 | 1,497 | 69.11 |
| Raw / Marginal (M) | 1,715 | 79.18 | 1,674 | 77.29 |
| Raw / Query (M) | 1,700 | 78.49 | 1,645 | 75.95 |
| Corrected / Frozen (M) | 1,737 | 80.19 | 1,371 | 63.30 |
| Corrected / Marginal (M) | 1,729 | 79.82 | 1,696 | 78.30 |
| Corrected / Query (M) | 1,740 | 80.33 | 1,704 | 78.67 |
| Corrected / improved update (T) | 1,770 | 81.72 | 1,730 | 79.87 |

The T update keeps the *same corrected prior as measured E4*; it does not use the E3 target prior. T gains over Query are +30/+26 correct calls (+1.39/+1.20 pp); gains over Marginal are +41/+34 (+1.89/+1.57 pp). Natural T stays below the fixed-menu oracle cap of $618 \times 3 = 1,854$. Each policy generates its own selected-feedback history.

**Decision test.** Require the 1 pp development resource line and nonnegative leave-one-fold-out contrasts within each fixed score scale. Cross-scale differences also change intervention exposure; they cannot isolate the update's benefit.

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## 4. Retain efficiency while adding useful capacity

E5 caps preserve room for an improved mechanism. They are engineering budgets, not predictions that every latency will decrease.

**Table 6:** E5 / body Table 7. Mean of stream-level p95s ($\mu$s); T rows are caps.

| | Natural | | Controlled | |
|---|---|---|---|---|
| Method / status | Selection | Update | Selection | Update |
| Frozen (M) | 15.89 | 1.31 | 17.88 | 1.50 |
| Online SGD (M) | 14.08 | 19.37 | 14.29 | 20.02 |
| Greedy LinUCB (M) | 128.63 | 90.90 | 128.36 | 92.32 |
| BayesianRouter (M) | 133.65 | 88.96 | 137.34 | 92.55 |
| Current MIRA-Relay (M) | 27.21 | 2.81 | 32.47 | 3.46 |
| Improved module (T; caps) | $\leq 30$ | $\leq 5$ | $\leq 35$ | $\leq 5$ |

macOS arm64; mean of 30 stream p95s, not pooled p95. Excludes encoding, provider forward, I/O and bookkeeping; one call/input. If the update cap is met, at least $19.37/5 = 3.87\times$ is the speedup over current natural SGD *for this kernel metric only*. Live memory and full-stack latency remain unmeasured.

**Decision test.** Meet efficacy and kernel budgets jointly. Measure encoding, queues/history, memory and recovery before deployment claims; keep label/call permissions matched.

## 5. Reduce accepted errors without reducing coverage

E6 is a secondary analysis of the inspected cohort: five fit-only policies, each applied to 157 molecules ($N = C = 785$ records/calls per policy). It is not new untouched confirmation.

**Table 7:** E6 / body Table 8. Matched-count refusal target; T is not an observed policy.

| Policy / status | Accepted | Wrong | Refused | Coverage | Acc.* | Utility |
|---|---|---|---|---|---|---|
| Always accept (M) | 785 | 93 | 0 | 100.00 | 88.15 | $-0.1285$ |
| Confidence refuse (M) | 663 | 36 | 122 | 84.46 | 94.57 | $-0.0947$ |
| Always refuse (M) | 0 | 0 | 785 | 0.00 | — | $-0.2600$ |
| Same-coverage random† (M) | 663 | 75 | 122 | 84.46 | 88.69 | $-0.1444$ |
| Improved ranking (T) | 663 | 28 | 122 | 84.46 | 95.78 | $-0.0845$ |

* Accepted accuracy; coverage and accuracy in %. † Evaluator-only conditional random null. $U = -(W + 0.25R + 0.01C)/N$, where $W$ is accepted errors and $R$ refusals. Always-refuse accuracy is undefined, not zero. T implies 635 correct accepted outputs and $U = -(28 + 0.25 \times 122 + 0.01 \times 785)/785 = -0.0845$.

**Decision test.** Freeze a fit-only ranking; report risk–coverage and cost sensitivity. T swaps eight accepted errors for eight correct outputs; the unchanged threshold cannot do this. Exact coverage is analytical, not guaranteed prospectively. Utility gains: +0.0102 over current confidence refusal; +0.0440 over always accept.

## 6. What would turn these goals into evidence?

1. Close the natural-selection and attribution gaps first. Isolate E3 prior and E4 update effects, then combine them under a matched budget to establish one coherent method.
2. Freeze the candidate, comparators and statistical plan before new evaluation. The inspected 157 molecules are not untouched confirmation; transfer target rates, not old counts.
3. Retain negative results and independent recovery. Endpoint, lifecycle and agent extensions need separate evidence. The full scientific gate remains **HOLD**.

**Sources (paths/hashes in target_spec.json):** E1/E5 fresh natural/stress; E2 mechanism v2; E3 headroom/learned-prior; E4 calibration v1; E6 corrected post-call audit v3. Real-paper tables remain unchanged.

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